

Project Overview & Key Specs

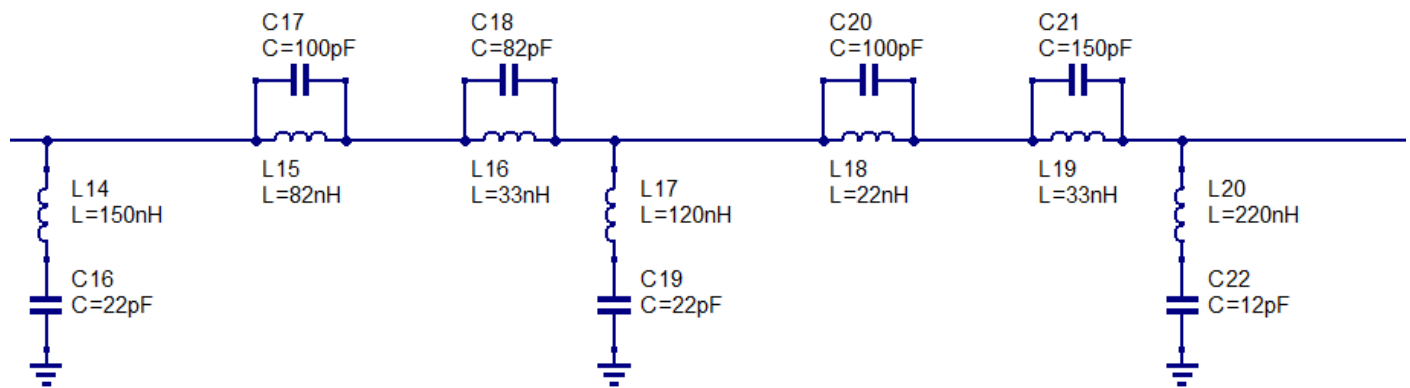
- **Target:** 137.5 MHz VHF Downlink
- **Active Device:** PGA-103+ E-PHEMT MMIC
- **Out-of-band Rejection (88–108 MHz):** >34 dB to -50 dB (Cascaded response)
- **Filter Insertion Loss (137 MHz):** 1.09 dB (Cauer Notch) / 2.5 dB (Butterworth Passband)
- **Stability:** Compensated unconditionally stable from 10 MHz to 4 GHz (R+L shunt & Π -resistor network)
- **PCB Layout:** CPGW (50 Ω), DC = PMOS + JST + Ferrite Bead

$W = 1.25\text{ mm} ; H = 1.6\text{ mm} ; S = 0.24\text{ mm} ; \epsilon_r = 4.5 ; t = 1\text{ oz (35}\mu\text{m)}$

Parameter	Target value	Simulated result
Operating Frequency	137.5 MHz (VHF Downlink)	137.5 MHz
Cascaded Gain (S_{21})	> 15 dB	19.2 dB
Noise Figure (NF_{total})	< 1.5 – 2.0 dB	1.84 dB
Input Return Loss (S_{11})	Noise-matched compromise (Γ_{opt})	-12.8 dB (a 137 MHz)
Out-of-band Rejection (FM)	> 10–15 dB (Input stage)	-34.8 to -50 dB (Cascaded)
Stability Range	Unconditional ($K > 1, \mu > 1$)	10 MHz to 4 GHz
DC Supply Voltage	4.8 V – 5.2 V	4.97 V (al pin V_{dd})
Linearity (IIP_3/OIP_3)	High dynamic range (FM saturation immunity)	$IIP_3 \approx +10.2\text{ dBm}$ $(OIP_3 \approx +36.7\text{ dBm})$

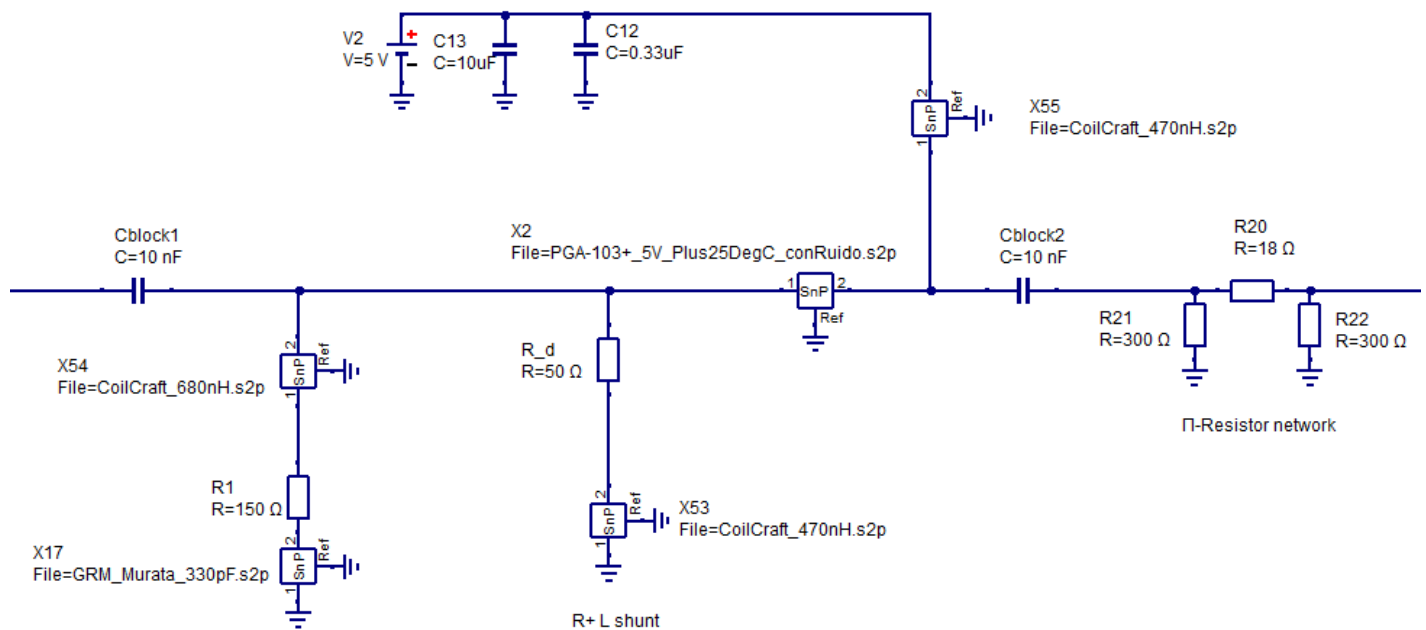
Schematics.

Cauer:

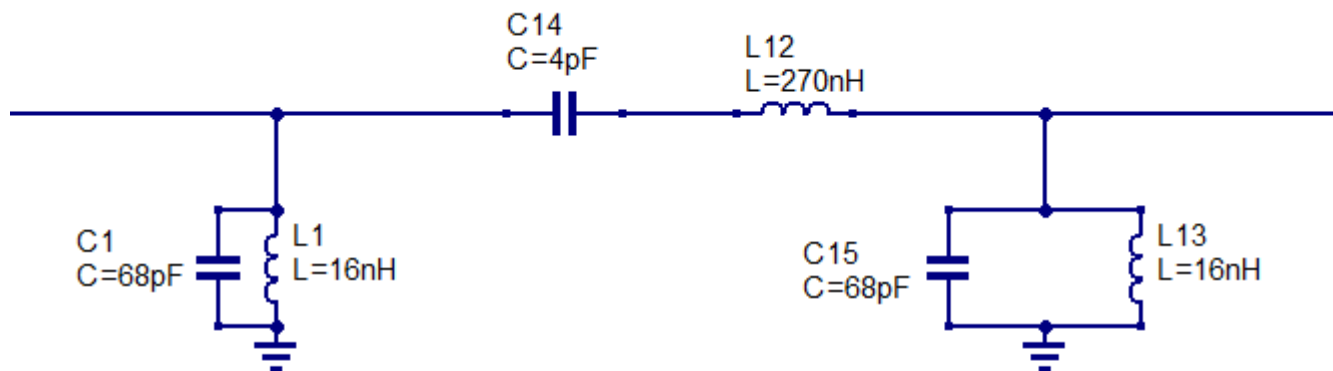


Cauer bandstop filter of order 5
96.7MHz cutoff/center frequency
impedance 50Ω

LNA:



Butterworth:



band-pass filter, 125MHz...170MHz
3th order Butterworth , PI-type,
impedance 50 Ω

PCB screenshot from KiCad:

